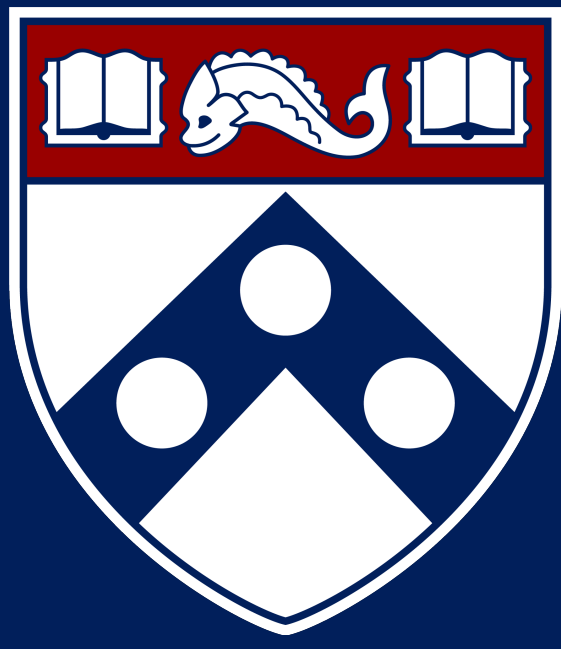


Why do marked words persist?



Corpus and experimental evidence for competition between phonological and semantic well-formedness in word formation and the lexicon

Po-Hsuan Huang

pohsuanh@sas.upenn.edu

Department of Linguistics, University of Pennsylvania

Canaan Breiss

cbreiss@uchicago.edu

Department of Linguistics, University of Chicago



Takeaway

A phonologically marked form survives in a language's lexicon and is tolerated in word formation when there are fewer semantically equivalent alternative forms.

Background

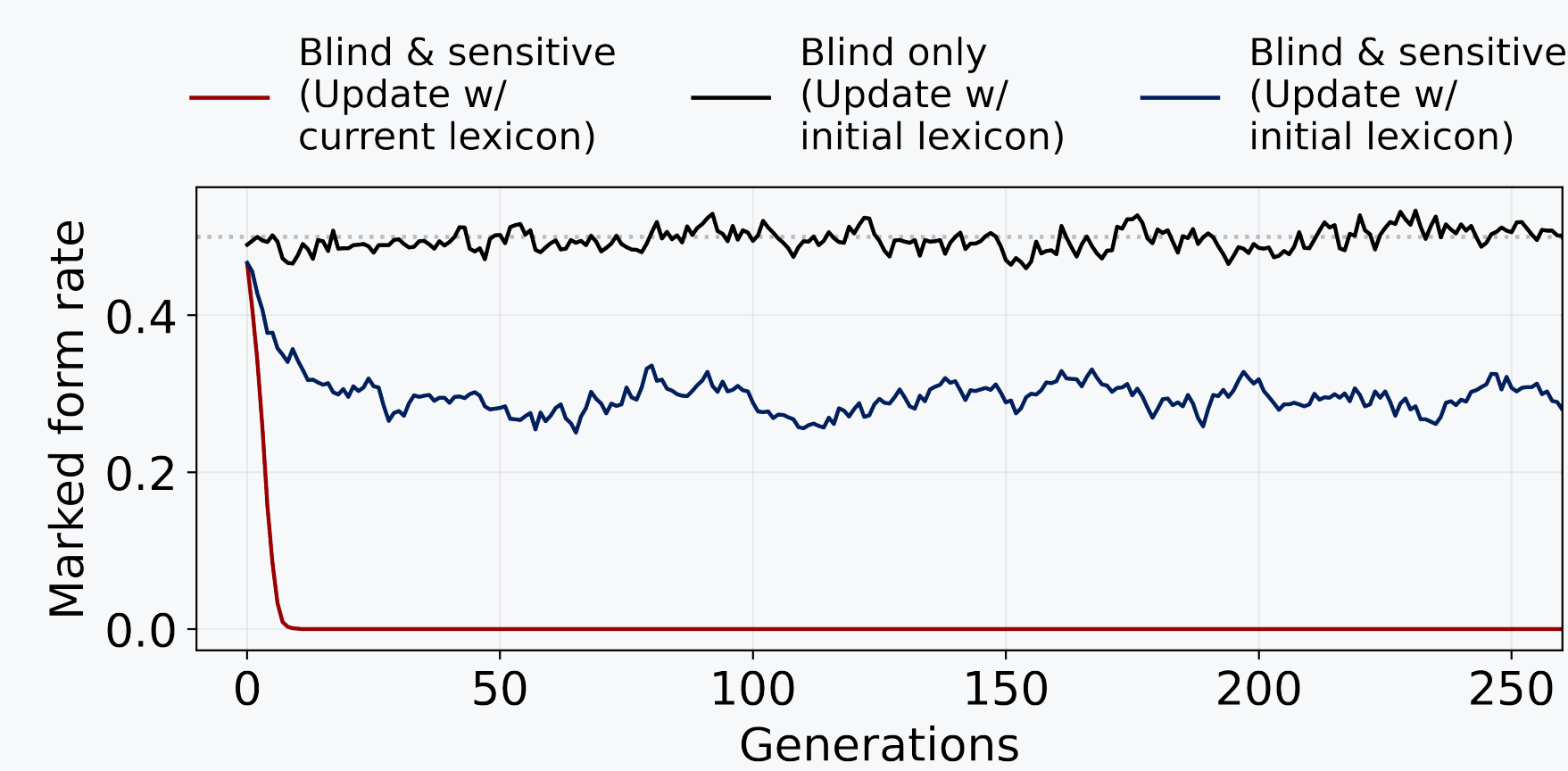
In probabilistic grammar, marked forms tend to become **marginalized** and die out over time.

Methods

MaxEnt models learned from the previous generation and produced new data for the following generation.

Table 1. Training data at initialization.

	CC	C+C
pt	100%	50%
pp	0%	50%



However, there are exceptions where marked forms are **tolerated** [1, 2, 3].

The mystery of survival: If marked forms are dispreferred, why do they survive in the lexicon?

Hypothesis

Word form flexibility (Flex) competes with phonological well-formedness.

Flex= N of alternative forms compatible with the target semantics.

Corpus studies

Language	Constraint	Example
English	*GEMINATE	<i>book. keep-er</i>
Finnish	*[a _{back}][–a _{back}]	<i>kesä.loma-a</i> /'kesæ.loma:/ "summer vacation"
Hungarian	*[a _{back}][–a _{back}]	<i>kő.por</i> /'kø:.por/ "rock flour"

Data two-member compounds from Wiktionary and published datasets [4, 5]

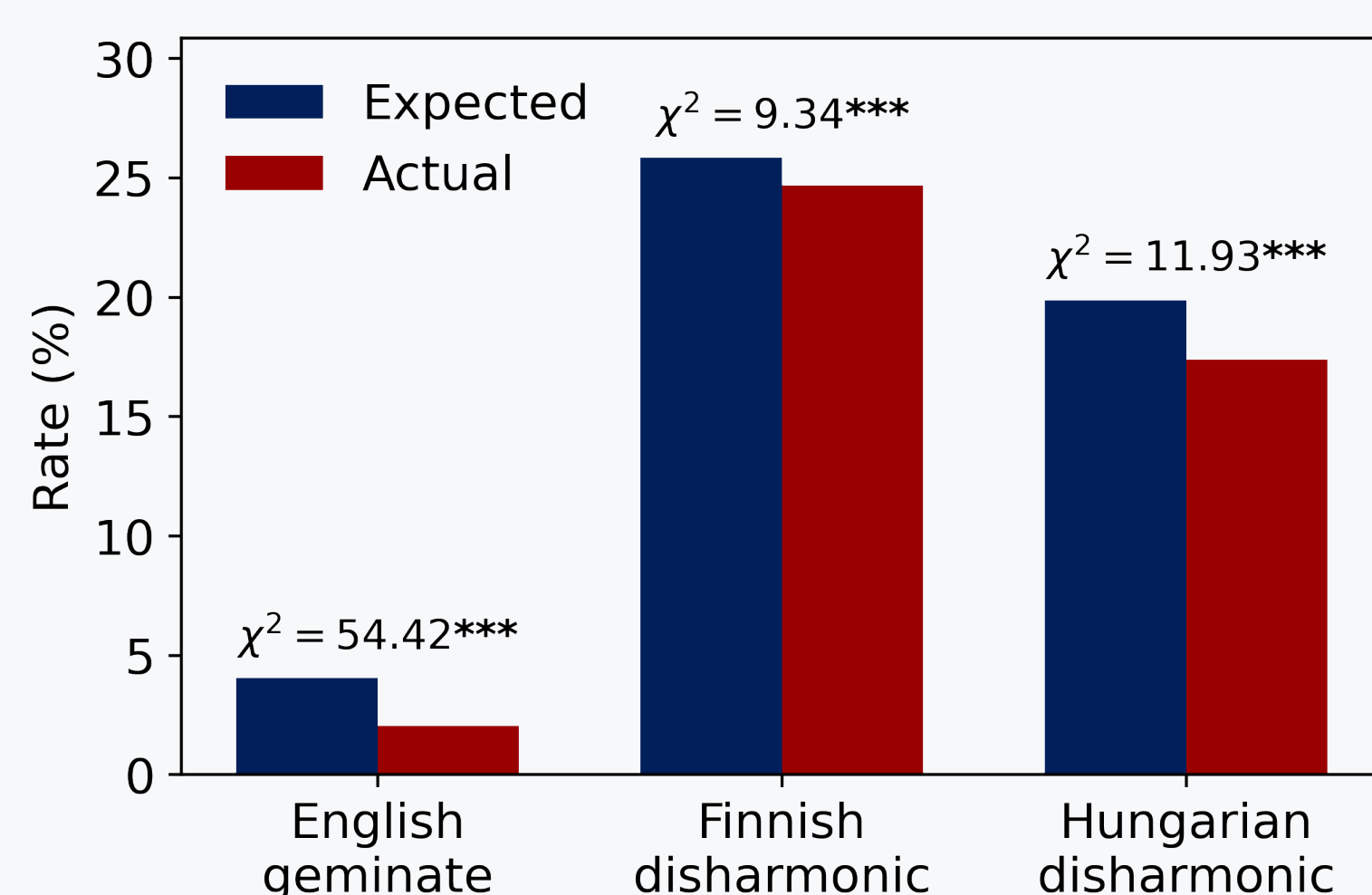
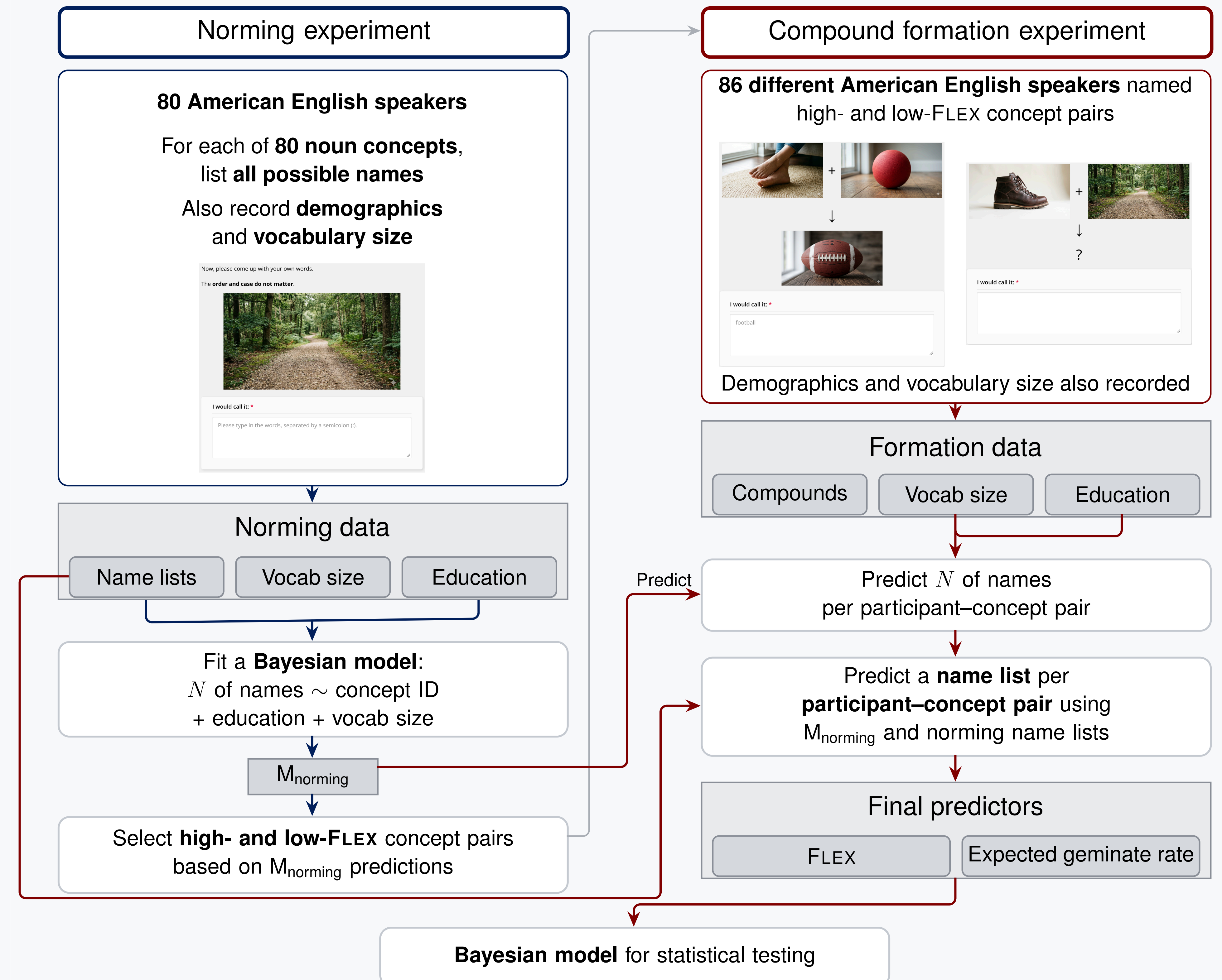


Figure 1. Geminate and disharmonic compounds are underrepresented.

Methods

- Flex(compound) = Flex(member1) × Flex(member2)
- Flex(word) = summed cosine similarity of the word's vector w/ all word vectors in the language

Compound formation experiment



Corpus results

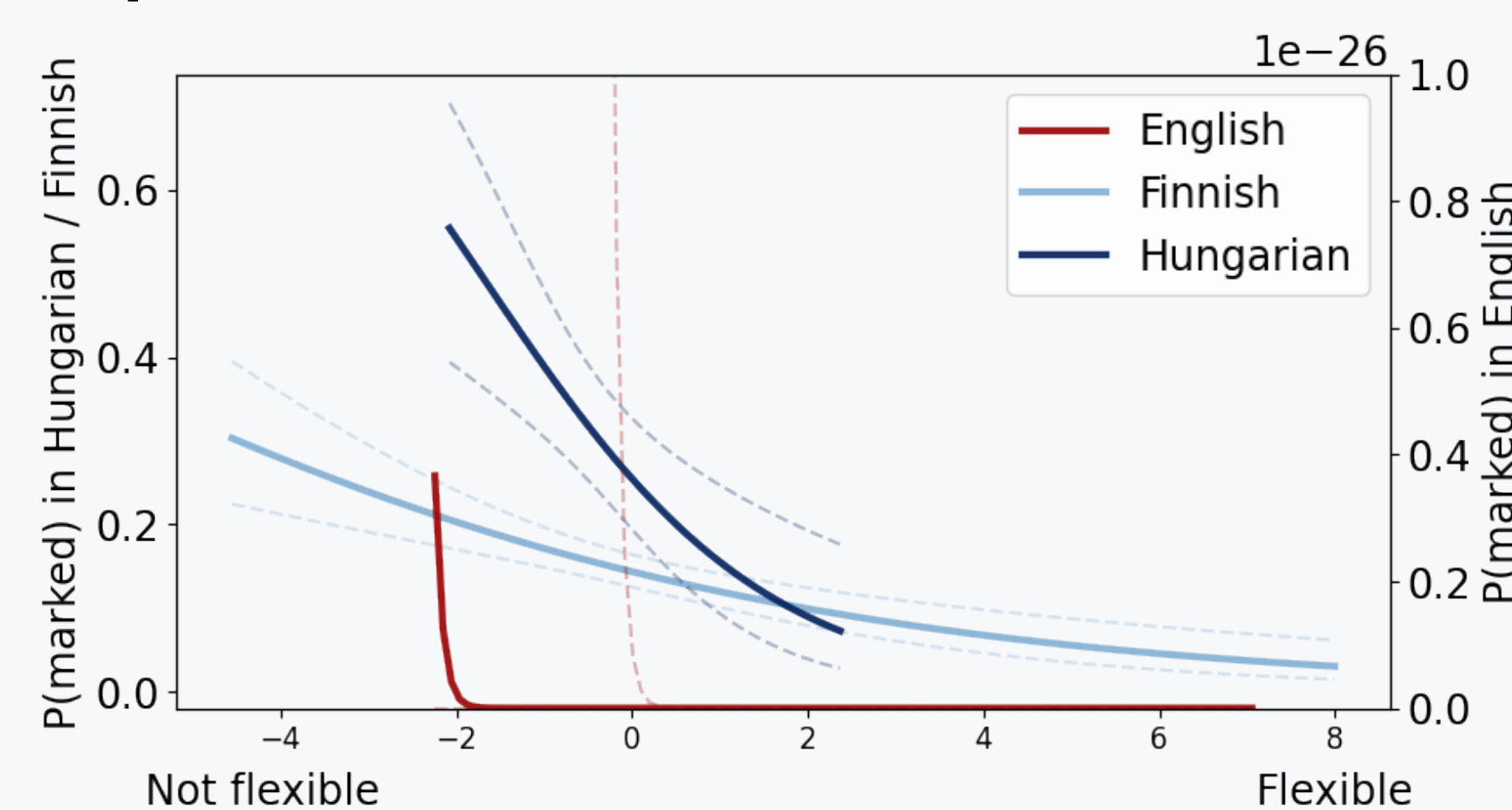


Figure 2. Flexeffects on a compound's markedness.

Examining diachronic evolution through MaxEnt simulation

Flex:

For each word:

- $k-1$ candidate forms were created, w/ the same geminate rates.
- k was randomly sampled from a Zipfian distribution (exponent= s).

Simulation results

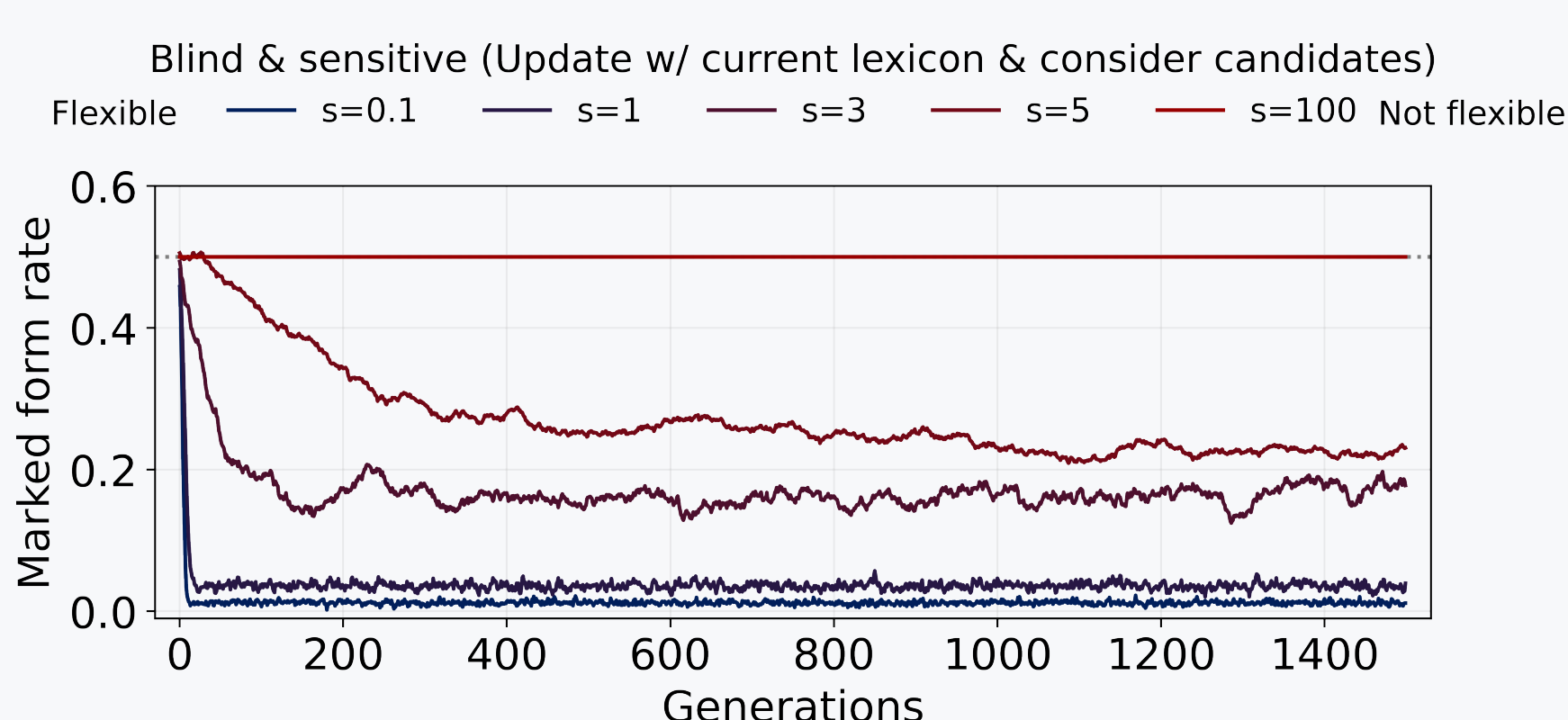


Figure 3. Simulated evolution of fake geminate compounds in the lexicon.

Experiment results

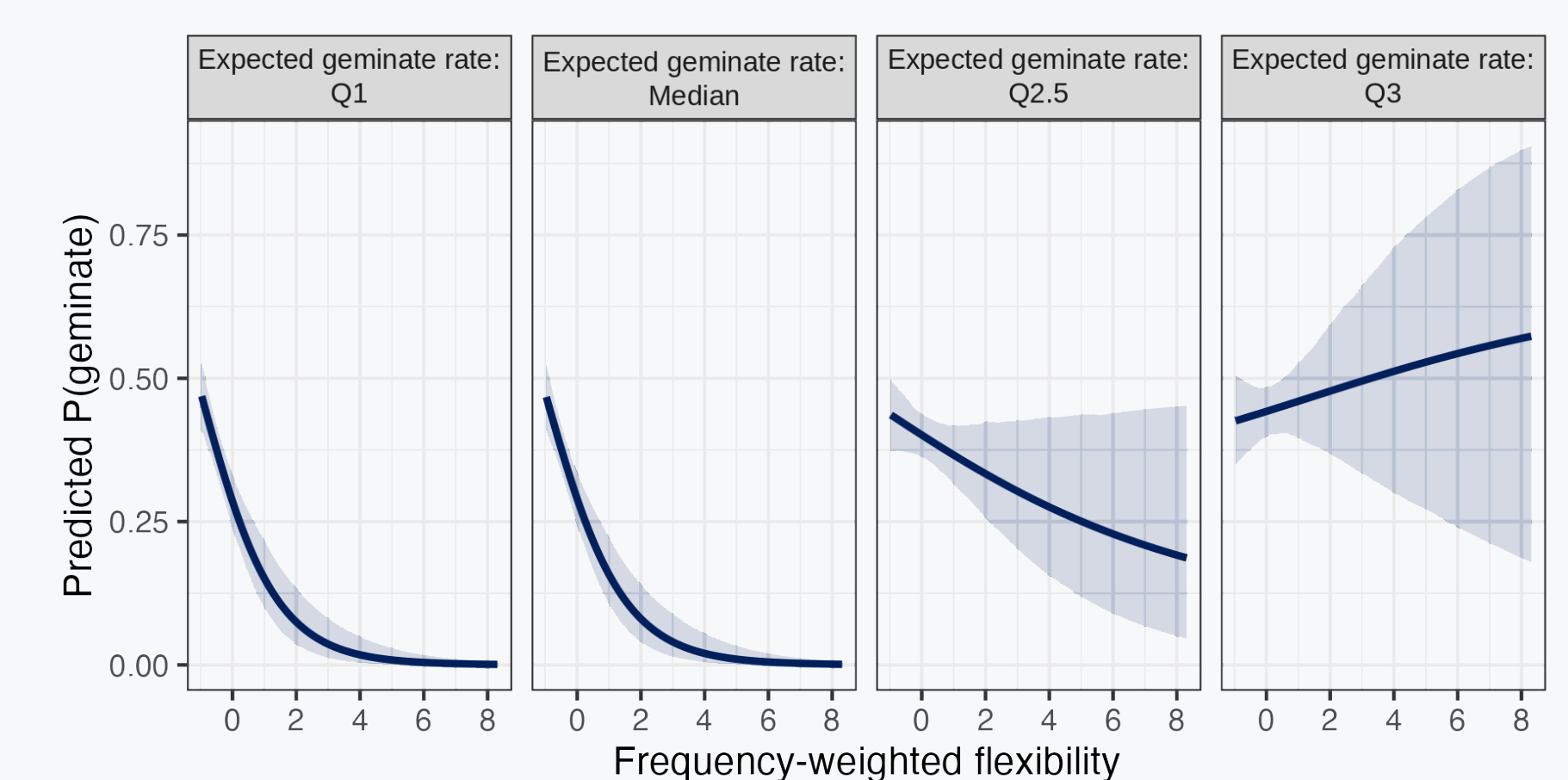


Figure 4. Model-derived interaction b/t a compound's Flex and the markedness rate of the name offered.

Selected references

- [1] Kempen, G., & Harbusch, K. (2008). Comparing linguistic judgments and corpus frequencies as windows on grammatical competence: A study of argument linearization in German clauses. In A. Steube (Ed.), *The discourse potential of underspecified structures* (pp. 179–192). De Gruyter.
- [2] Martin, A. (2011). Grammars leak: Modeling how phonotactic generalizations interact within the grammar. *Language*, 87(4), 751–770.
- [3] Breiss, C., & Hayes, B. (2020). Phonological markedness effects in sentence formation. *Language*, 96(2), 338–370.
- [4] Gagné, C. L., Spalding, T. L., & Schmidtke, D. (2019). LADEC: The Large Database of English Compounds. *Behavior Research Methods*, 51, 2152–2179.
- [5] Günther, F., & Marelli, M. (2019). Enter sandman: Compound processing and semantic transparency in a compositional perspective. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 45(10), 1872–1882.

Acknowledgment

We thank Yue Chen and Linh Pham for providing directions during the design of the compound formation experiment and the PhonLunch@USC audience for their feedback on an earlier version of this study.



Poster PDF